

British Airways

Task 2

For Task 2, BA wants you to use data + machine learning to answer:

Can we use customer data to guess if a customer will complete a booking or not?

Example

Suppose we know:

Customer Information	Value
Purchase Lead	60 days
Route	London → Singapore
Length of Stay	10 days
Flight Duration	13 hours

Using this information, the machine learning model predicts:

Booking Complete = 1

Meaning:

✅ Customer is likely to complete the booking.

Another customer:

Customer Information	Value
Purchase Lead	2 days
Length of Stay	1 day
Route	Short route

Model predicts:

Booking Complete = 0

Meaning:

✗ Customer is unlikely to complete the booking.

Why BA wants this

If BA can predict who is likely to book:

- They can send special offers to customers who are unlikely to book.
- They can target marketing campaigns more effectively.
- They can increase sales and revenue.

Now you need to do **3 things**:

1. Prepare the data

2. Train a model

3. Create 1 PowerPoint slide with findings

Step 1: Convert Text to Numbers

Code

```
from sklearn.preprocessing import LabelEncoder
```

```
le = LabelEncoder()
```

```
for col in ['sales_channel','trip_type','route','booking_origin']:  
    df[col] = le.fit_transform(df[col])
```

Why?

Your dataset contains text:

sales_channel

Internet

Mobile

Machine learning models cannot understand words.

They only understand numbers.

So LabelEncoder converts:

Before	After
Internet	0
Mobile	1

Now the model can use these columns.

Step 2: Separate Features and Target

Code

```
X = df.drop("booking_complete", axis=1)
y = df["booking_complete"]
```

Why?

The model needs:

X = Inputs

Things we know about the customer:

- purchase_lead
- flight_duration
- trip_type
- route
- booking_origin

y = Output

What we want to predict:

booking_complete

booking_complete

1

0

1

Meaning:

- 1 = Customer booked
 - 0 = Customer didn't book
-

Step 3: Split Data

Code

```
from sklearn.model_selection import train_test_split
```

```
X_train, X_test, y_train, y_test = train_test_split(  
    X,  
    y,  
    test_size=0.2,  
    random_state=42  
)
```

Why?

We don't want the model to memorize all data.

We split:

Dataset Purpose

80% Train model

20% Test model

Example:

50,000 rows

- 40,000 for learning
 - 10,000 for testing
-

Step 4: Train Random Forest

Code

```
from sklearn.ensemble import RandomForestClassifier
```

```
model = RandomForestClassifier(  
    n_estimators=100,
```

```
random_state=42
)

model.fit(X_train, y_train)
```

Why?

Random Forest creates many decision trees.

Example tree:

Purchase Lead > 30 days?

|

Yes

|

Booked before?

|

Yes

|

Likely to book

100 trees vote together.

This usually gives good predictions.

Step 5: Predict

Code

```
y_pred = model.predict(X_test)
```

Why?

Now the model makes predictions.

Example:

Actual Predicted

1 1

0 0

1 0

The model is guessing whether customers will book.

Step 6: Accuracy

Code

```
from sklearn.metrics import accuracy_score
```

```
accuracy = accuracy_score(y_test, y_pred)
```

```
print(accuracy)
```

Why?

Accuracy answers:

How often was the model correct?

Example:

10,000 test records

Model gets 8,500 correct

8500 / 10000

=

85%

Accuracy = 0.85

Step 7: Classification Report

Code

```
from sklearn.metrics import classification_report
```

```
print(classification_report(y_test, y_pred))
```

Why?

Accuracy alone can be misleading.

This report gives:

Precision

When model predicts booking:

How often is it right?

Recall

Of all customers who booked:

How many did we find?

F1 Score

Balance of Precision and Recall.

BA likes seeing these metrics.

Step 8: Cross Validation

Code

```
from sklearn.model_selection import cross_val_score
```

```
scores = cross_val_score(  
    model,  
    X,  
    y,  
    cv=5  
)
```

```
print(scores.mean())
```

Why?

Instead of testing once:

The model tests itself 5 times.

Example:

Fold 1 = 84%

Fold 2 = 85%

Fold 3 = 83%

Fold 4 = 86%

Fold 5 = 84%

Average:

84.4%

More reliable than one test.

Step 9: Feature Importance

Code

```
importance = pd.DataFrame({  
    'Feature': X.columns,  
    'Importance': model.feature_importances_  
})  
  
importance = importance.sort_values(  
    by='Importance',  
    ascending=False  
)  
  
print(importance)
```

Why?

This is the most important part for BA.

It answers:

Which factors influence bookings the most?

Example:

Feature	Importance
purchase_lead	0.25
flight_duration	0.18
route	0.15
booking_origin	0.12

Interpretation:

Purchase Lead is the strongest predictor of booking behaviour.

Step 10: Visualization

Code

```
importance.plot(  
    x='Feature',  
    y='Importance',  
    kind='bar'  
)
```


Why?

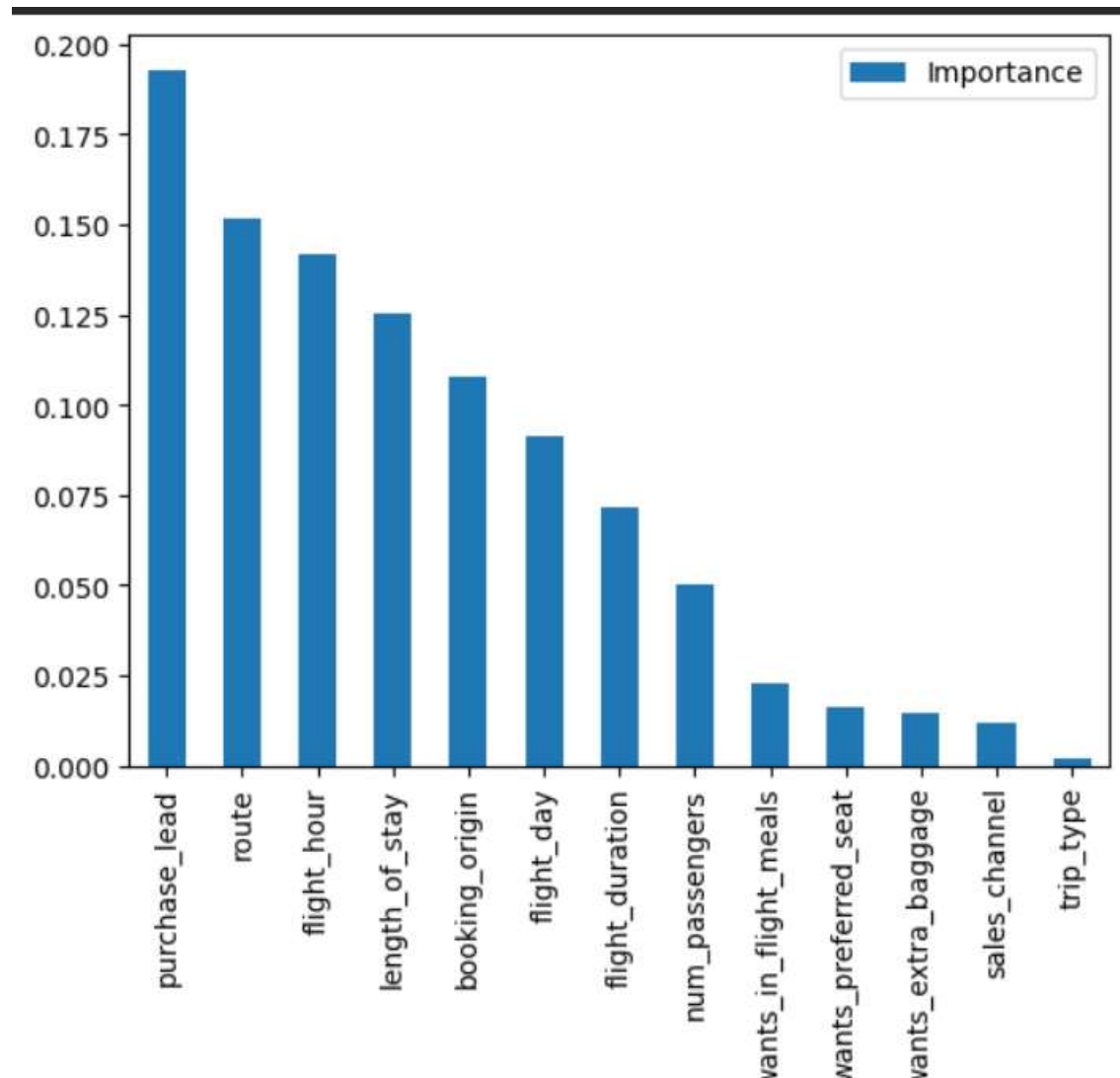
Managers prefer charts.

Instead of:

`purchase_lead = 0.25`

`route = 0.15`

They see a clear bar chart showing the most important variables.



Key Drivers of Booking Completion (Easy Explanation)

The chart shows which factors have the biggest impact on whether a customer completes a booking.

- **Purchase Lead** was the most important factor. This means customers who book their flights well in advance are more likely to complete their booking.

- **Route** was the second most important factor. Some flight routes are more popular and have higher booking completion rates than others.
- **Flight Hour** also played an important role. The time of departure can influence a customer's decision to book.
- **Length of Stay** was another key factor. The number of days a customer plans to stay at their destination affects booking behaviour.
- **Booking Origin** was important as well. Customers from different countries may have different booking patterns and preferences.

Less important factors included:

- Extra baggage
- Preferred seat
- In-flight meals
- Sales channel

The **least important factor** was **Trip Type** (Round Trip, One Way, or Circle Trip), which had very little impact on booking completion.

Conclusion

The analysis shows that customers who book earlier, travel on certain routes, and fly at specific times are more likely to complete their booking. These insights can help British Airways target customers more effectively and improve marketing strategies.

Note: google colab notebook link

https://colab.research.google.com/drive/1sEZ83a0ZMcH0v_pU3Aj8u4K7tO8ihq-T?usp=sharing